

TrES-2b

R-2242

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_110530 · created 2026-08-15T23:10:26+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2455711.770885 +/- 8.4e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.11 +/- 0.016
Transit depth (Rp/Rs)^2	1.22 +/- 0.35 [%]
Orbital Inclination (inc)	84.24 +/- 0.59
Airmass coefficient 1 (a1)	1.0033 +/- 0.0082
Airmass coefficient 2 (a2)	-0.0023 +/- 0.0047
Scatter in the residuals of the lightcurve fit is	0.82 %
Best Comparison Star	#2 - [258, 436]
Optimal Aperture	4.55
Optimal Annulus	6.28
Transit Duration (day)	0.0771 +/- 0.009

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.11 ± 0.016 vs 0.1254 (deviation 0.96σ)
Duration fit vs published	0.0771 d vs 0.0746 d (deviation 0.28σ)
Mid-time O–C	-0.2 min
Depth SNR	6.9
Residual scatter	0.82 %

Light curve

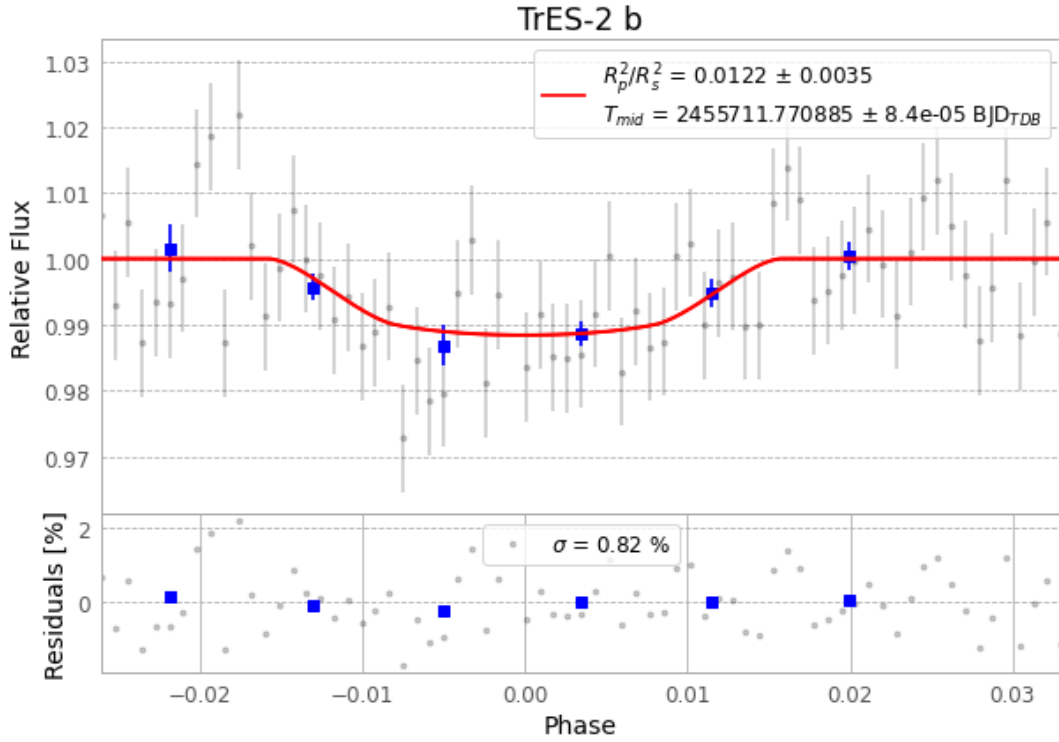


Figure 1 — FinalLightCurve_TrES-2 b_2011-05-29.png (SHA-256 c29d09a20c7d4d73...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [322, 235], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 8914bda03de4d6b2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

71 FITS frames (47 MB), TRES-2110530045413.FITS → TRES-2110530082412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-110530.log (71ede2aaf73c...), FinalLightCurve_TrES-2 b_2011-05-29.png (c29d09a20c7d...), FinalLightCurve_TrES-2 b_2011-05-29.csv (672e662720ba...)

Compute resources

Wall time 112.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-15 23:10Z.